SIH26186 • MEDTECH / SECURITY

# VIGIL-AI

## Predictive Force Welfare & Stress Triage Portal

### BULBA BYTE BENDERS

- AI/ML Modeling
- Flutter Frontend
- Wearable BLE Data
- PostgreSQL Database
- Privacy & Security API
- Deployment & Docs



# Team Member Details

| Member | Name | Role / Contribution |
|---|---|---|
| Team Leader | Vanshika Nakra | Project Idea, Maximising team output, Product owner and visionary, Pitch strategy |
| Member 2 | Sarthak Chaudhari | UI/UX Designer, Rule based logic system, Score algorithm |
| Member 3 | Darshil Gehlot | APIs and RLS(Row Level Security) |
| Member 4 | Nikhil Jangid | SupaBase, Database integration, Realtime data processing and Researcher |
| Member 5 | Himanshu Saxena | Intelligence support system |
| Member 6 | Md Shahid Afridi | AI Integration |

THE OPERATIONAL REALITY

# Data Silos & Late Support

High Workload → Low Recovery → Accumulated Silent Stress → Late Support

Currently, critical wellbeing data is trapped in isolated silos:

### Operations
Shifts, tasks, and workload.

### Wellbeing
Recovery, stress, and check-ins.

### Human Support
Medics, supervisors, and buddies.

BRIDGING THE GAP

# Proactive Triage & Explainable AI

VIGIL-AI actively connects operational workload, shifts, and recovery patterns to provide contextual insights and trigger early human support.

### 14-Day Baseline

Replaces generic thresholds. Learns specific personal baselines to prevent false alarms.

### Explainable AI (XAI)

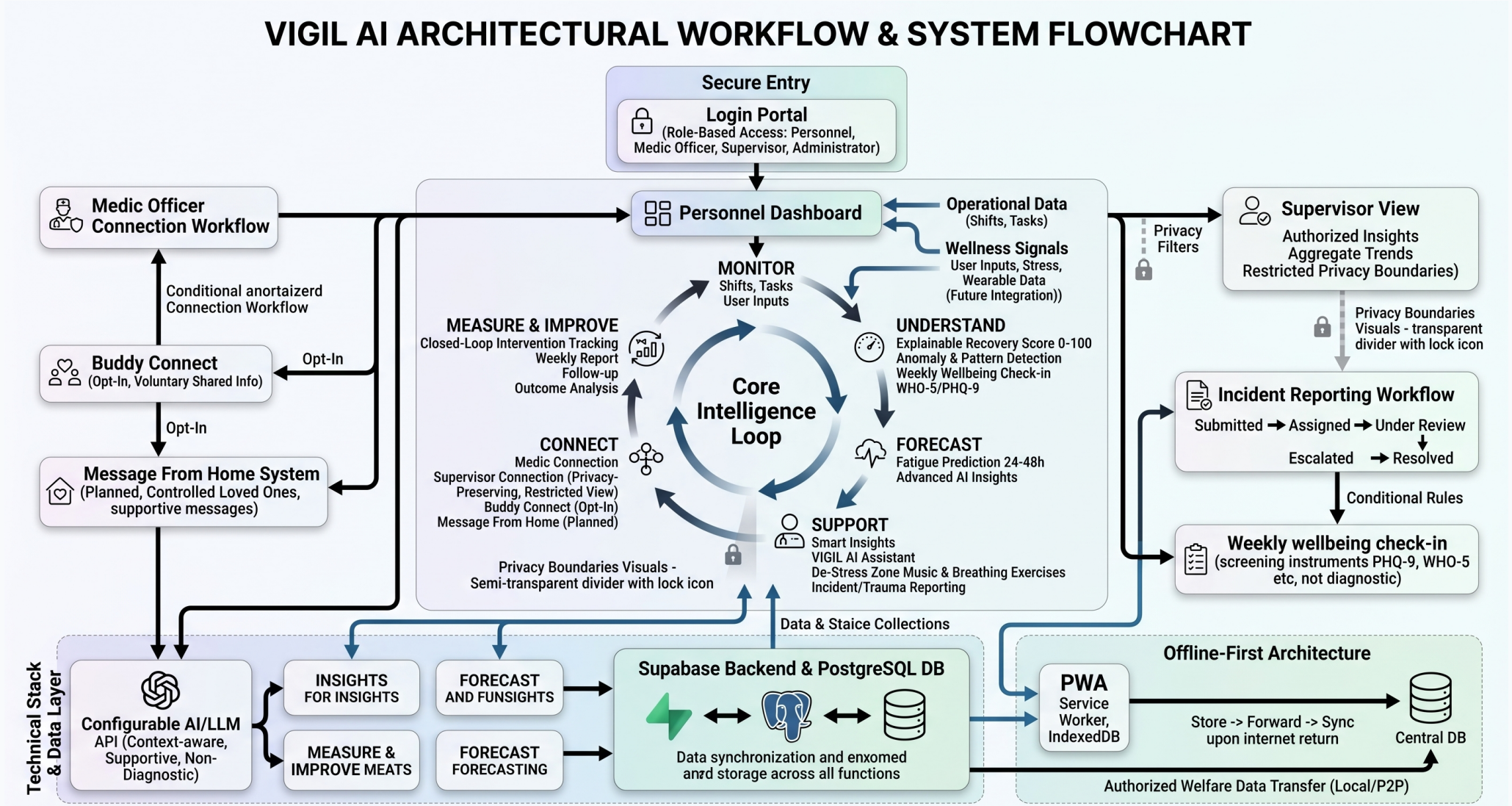
Differentiates normal physical exertion from actual psychological distress.

### Proactive Triage

Acts as an early-warning system. Triggers interventions before burnout occurs.

SYSTEM ARCHITECTURE

# Workflow & Core Stack



CONTINUOUS SUPPORT LOOP

# Explainable Intelligence Metrics

0-100

## Recovery Score

Driven by sleep, physiological signals, and workload against the 14-day baseline.



## Stress Load Score

Contextualized by operational shifts, trauma incidents, and WHO-5 check-ins.

24-48h

## Fatigue Risk

4-tier risk classification forecasting cumulative operational strain.

DPDP ACT 2023 COMPLIANT

# Secure, 4-Role Ecosystem

Strictly role-based access to reduce mental health stigma and ensure absolute privacy.



## Personnel

Complete ownership of personal wellbeing context, insights, and support tools.



## Supervisor

Views anonymized unit heatmaps for strategic rest-rotation decisions.



## Medic

Reviews authorized wellbeing signals and interventions for medical escalation.



## Administrator

Handles verification, security architecture, and organization-level oversight.

EMPOWERING THE SOLDIER

# Integrated De-escalation

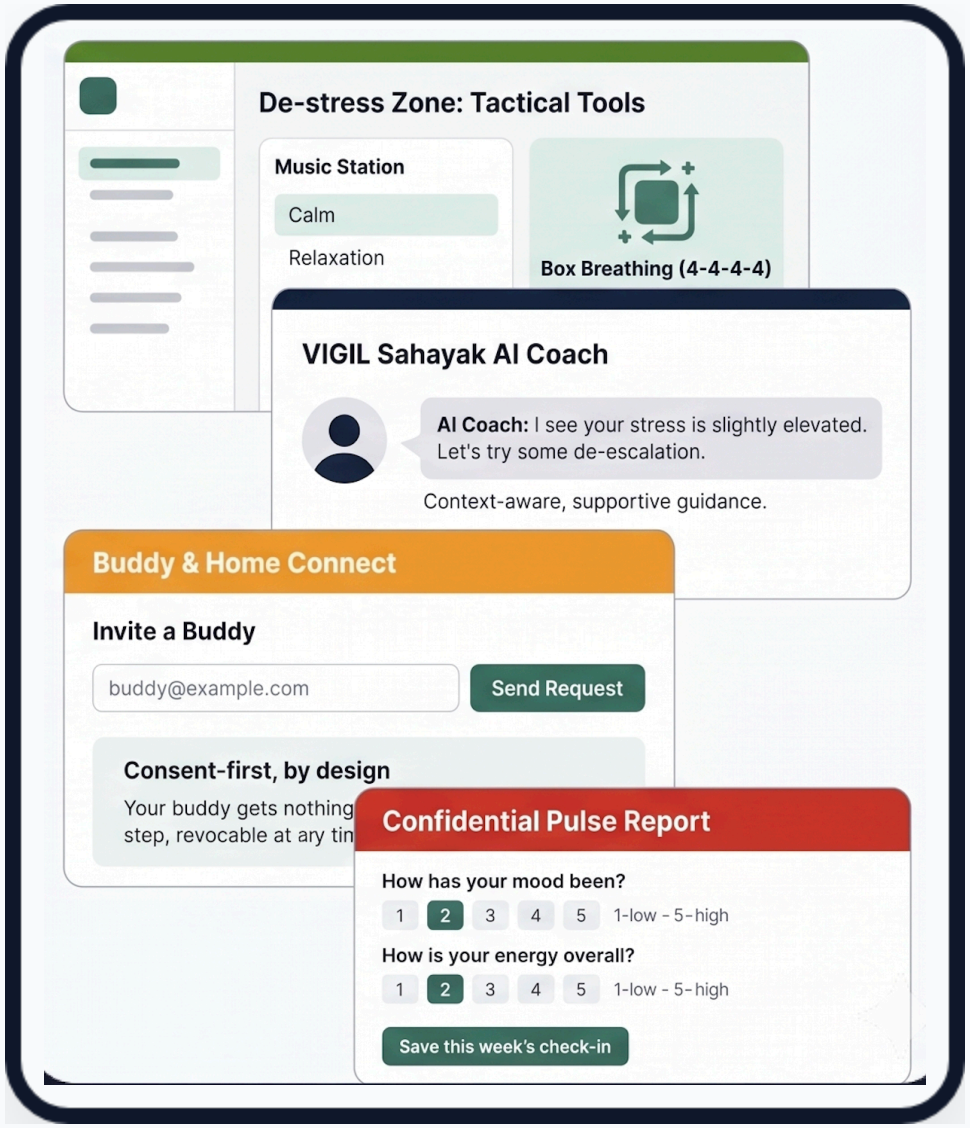
Designed to answer: "What is happening with me, and what can I do next?"

### Tactical Relief Tools
Offline-capable 4-4-4-4 Box Breathing and Sensory Grounding.

### AI Sahayak Coach
Context-aware, supportive, non-diagnostic guidance.

### Buddy & Home Connect
Opt-in peer support and controlled messages from loved ones.

### Confidential Check-ins
60-second operational pulse reports linked to Tele-MANAS (14416).



REAL-WORLD EXECUTION

# Feasibility, Mitigation & Scale

### Data Security & Stigma

Zero unconsented sharing. Client-side encryption. Framed as "autonomic recovery" to remove mental illness stigma.

### Wearable Noise

14-day personal rolling baseline absorbs device noise. Multi-factor AI verifies biometrics against manual checks.

### Future Expansion

- Broader COTS wearable integration.
- Predictive trauma models.
- Offline-first PWA synchronization.

# Protecting Those Who Protect Us

VIGIL-AI transitions force welfare from reactive crisis management to proactive early detection.



### Operational Impact

Optimizes rest-rotation. Prevents fatigue-related tactical errors before they occur.



### Social Impact

Normalizes mental welfare. Reduces help-seeking stigma. Provides immediate trauma grounding.



### Economic Impact

Scalable via COTS wearables. Drastically reduces costs tied to attrition and accidents.

MONITOR UNDERSTAND FORECAST SUPPORT MEASURE IMPROVE